

SwitchFiber ISP Platform Strategy

Industry-Standard Gap Analysis & Multi-Platform Architecture Breakdown

Ecosystem: Admin Console (Vue) • Tech Mobile (Flutter) • Subscriber Portal (Vue) • Core OSS/BSS

Executive Summary & Target Architecture

This document outlines modern industry standards for Fiber Internet Service Providers (ISPs) benchmarked against leading platforms (Splynx, Sonar, Ubiquiti UISP, and MikroTik RouterOS). The analysis is divided across our 3 active repositories and core network automation layers: (1) Admin Console (switchfiber), (2) Technician Mobile App (switchfibe_tech in Flutter), (3) Subscriber Self-Care Portal (switchfiberusers in Vue), and (4) Backend OSS/BSS Network Automation.

1. Admin Console (switchfiber — Vue 3 / PrimeVue)

Domain Area	Current Capabilities	Industry-Standard Missing Features	Priority & ROI
NOC & Optical Telemetry	Port tracking, LCP/NAP maps, status filters, basic RADIUS users.	- Live Optical Power (Rx/Tx dBm) telemetry from OLT/ONUs. - Optical Degradation Alerts (Yellow: -25 to -27 dBm, Red: < -28 dBm). - Mass Outage / Fiber Cut auto-detection by clustering down ports.	HIGH Reduces intermittent complaints by 70%.
Automated Billing & Dunning	Invoice creation, payment status dropdown, statement generation.	- Automated Dunning Lifecycle (Grace period -> Reminder -> Walled Garden). - Walled Garden / Captive Portal redirect on MikroTik rather than hard drop. - Dynamic QR Ph / GCash invoice generation.	CRITICAL Eliminates manual payment tracking & bad debt.
SLA Helpdesk & Ticketing	Service Orders for repairs and technician dispatches.	- Customer Trouble Ticketing with SLA countdown timers. - Automated ticket assignment by barangay / LCP cluster. - Incident Management with mass-broadcast SMS/Viber to affected NAPs.	HIGH Ensures SLA adherence and customer retention.
IPAM & Bandwidth Monitoring	Static IP column and router model definitions.	- Subnet Pool Manager (IPv4 CGNAT, Corporate Static Pools, IPv6 /56 prefixes). - Embedded live MRTG/Grafana bandwidth traffic graphs per subscriber. - Live MikroTik queue & bandwidth profile syncing.	MEDIUM Prevents IP conflicts & verifies bandwidth.
Inventory & Asset Tracker	Item quantities 1-10 on Job Orders.	- Centralized Warehouse Inventory (Fiber drop cables, ONUs, patch cords). - Auto-deduct stock upon Job Order completion. - Serial number tracking from warehouse to customer home.	MEDIUM Stops equipment leakage and loss.

2. Field Technician Mobile App (switchfibe_tech — Flutter)

Mobile Module	Current Capabilities	Industry-Standard Missing Features	Technical Implementation
Optical Signal Reading (OPM)	Box reading and router reading image upload.	- Digital Optical Power Meter (OPM) reading input with auto-validation. - Real-time signal pass/fail indicator (Green: -15 to -24 dBm). - Prevents completing job order if signal is worse than -27 dBm.	Flutter Form Validation & Signal Threshold Evaluator
Barcode & QR Hardware Scan	Manual text typing of modemRouterSN and MAC.	- Camera barcode scanner for instant Serial Number & MAC capture. - Prevents typo mistakes in PPPoE and RADIUS bindings. 1-tap scan to bind ONU directly from box barcode.	mobile_scanner or qr_code_scanner package
In-App Speed & Latency Test	Manual speedtest screenshot upload.	- Integrated in-app speed test engine. - Auto-records Download (Mbps), Upload (Mbps), Ping (ms), and Jitter. - Cryptographically signs test results with timestamp and GPS coordinates.	Custom HTTP multi-stream or Ookla Speedtest SDK
Offline Caching & Local Sync	Online network requests.	- Offline Job Order caching for zero-signal basement / remote areas. - Offline draft storage with auto-sync when 4G/Wi-Fi re-establishes. - Background queue with retry mechanisms for photo uploads.	Hive / SQLite (sqflite) + Background Sync WorkManager
Digital E-Signature & Turnover	Client signature image.	- Interactive smooth finger-signature pad on tablet/phone. - Digital Turnover Certificate generation (PDF) emailed to client upon sign-off. - Customer satisfaction star rating (1-5 stars) at completion.	signature package + in-app PDF rendering
Turn-by-Turn Fiber Navigation	Google Maps directions link.	- In-app routing to designated NAP pole coordinates. - Shows nearest available NAP box with free ports if assigned port is defective.	flutter_map / Google Maps SDK + GPS location stream

3. Subscriber Self-Care Portal (switchfiberusers — Vue)

Portal Feature	Standard ISP Scope	Industry-Standard Missing Features	Subscriber Impact
1-Click QR Payment & Webhooks	Manual bank transfer / over-the-counter payments.	- Instant checkout with GCash, Maya, ShopeePay, Credit Card, 7-Eleven. - Dynamic QR Ph on invoice. - Real-time webhook auto-unblocks internet within 60 seconds.	CRITICAL Reduces billing customer inquiries by 80%.
Live Connection Health Check	Subscriber calls hotline when offline.	"Is My Fiber Online?" diagnostic widget on dashboard. - Shows live ONU status (Online, Red LOS, Degraded) and Optical Power. - Displays ongoing maintenance/outage banners in subscriber's barangay.	HIGH Eliminates redundant support calls during outages.
Remote Wi-Fi CPE Management	Technician required to change Wi-Fi password.	- Self-service Wi-Fi SSID and password update via TR-069 ACS integration. - View connected devices / Wi-Fi signal strength meter. - One-click router remote reboot button.	HIGH Provides modern smart-home fiber experience.
Invoices & Statement of Account	Paper bills or emailed PDFs.	- Comprehensive billing history with downloadable PDF Official Receipts. - Automated email and SMS notifications when new invoice is generated. - Proof of Payment upload with instant AI/OCR reference verification.	MEDIUM Complies with tax and receipting standards.
Self-Service Ticket Submission	Reporting issues via social media chat.	- In-app ticket submission with photo upload (e.g. Red LOS flashing light). - Real-time technician dispatch tracker (En route, Arrived, Resolved). - Speed test history log.	HIGH Builds trust and transparency.

4. Core Network & Backend Integrations (OSS / BSS Layer)

Integration Layer	Industry Protocol / Tool	Functional Description	Business & Technical Value
OLT / PON Provisioning Engine	SNMP / Telnet / CLI / Netconf (Huawei, ZTE, VSOL, Fiberhome, Ubiquiti)	Automates ONU authorization on GPON/EPON OLTs. Binds serial number, applies VLAN, assigns line profile, and reads optical Rx/Tx power levels automatically.	Zero-touch provisioning: field tech connects ONU, system activates line in seconds.
RADIUS & MikroTik RouterOS	FreeRADIUS + RouterOS API / CoA (Change of Authorization)	Manages PPPoE sessions, speed limits (queues), and dynamic disconnection/reconnection. Enables Walled Garden redirect for unpaid subscribers.	Instant bandwidth throttling and real-time auto-reconnection upon payment.
TR-069 / TR-369 (USP) ACS	GenieACS / OpenACS / CWMP (HTTP/SOAP/WebSockets)	Auto Configuration Server (ACS) for remote Customer Premises Equipment (CPE). Pushes Wi-Fi configurations, firmware updates, and gathers diagnostics remotely.	Eliminates truck rolls for minor Wi-Fi setup or password change requests.
Payment Gateway Aggregators	PayMongo / Xendit / Dragonpay / Maya API (REST Webhooks + QR Ph)	Automated checkout links and webhook listeners that reconcile GCash, Maya, and bank transfers immediately into subscriber ledger.	100% automated collections with zero manual receipt auditing required.
Omnichannel Notification Gateway	Semaphore / Twilio / Viber Business API (HTTP Webhooks)	Automated multi-channel notifications for billing reminders (3 days before due date, on due date), maintenance advisories, and ticket updates.	Drastically lowers default rates and ensures 98%+ delivery rate.

5. Strategic Implementation Roadmap & Milestones

Phase	Timeframe	Focus Deliverables	Target Repositories	Key Performance Metrics (KPI)
Phase 1: Immediate Essentials	Month 1 - 2	- Payment Gateway Webhooks (GCash/Maya). - Automated SMS/Viber Billing Reminders. - QR/Barcode scanner in Flutter tech app. - Optical dBm signal validation fields.	switchfiber switchfibe_tech switchfiberusers	80% automated payment reconciliation. 0 barcode/SN typo errors in Job Orders.
Phase 2: Operational Excellence	Month 3 - 4	- SLA Helpdesk & Ticketing System. - Walled Garden MikroTik auto-quarantine. - Correlated Mass Outage / Fiber Cut alerts. - Offline sync & Speedtest in tech mobile app.	switchfiber switchfibe_tech Backend API	- Average resolution time < 4 hours. 0 lost offline job orders in field.
Phase 3: Advanced Automation	Month 5 - 6	- OLT 1-Click Auto-Provisioning. - TR-069 Remote Wi-Fi & Router Reboot. - Subscriber Self-Care live line diagnostics. - Live SNMP Bandwidth Graphs.	switchfiber switchfiberusers GenieACS / OLT	60% reduction in customer support hotline load. 5-minute zero-touch customer activation.